

Question ID 29c177e6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 29c177e6

SAT3 M 2.1

What is 10% of 470?

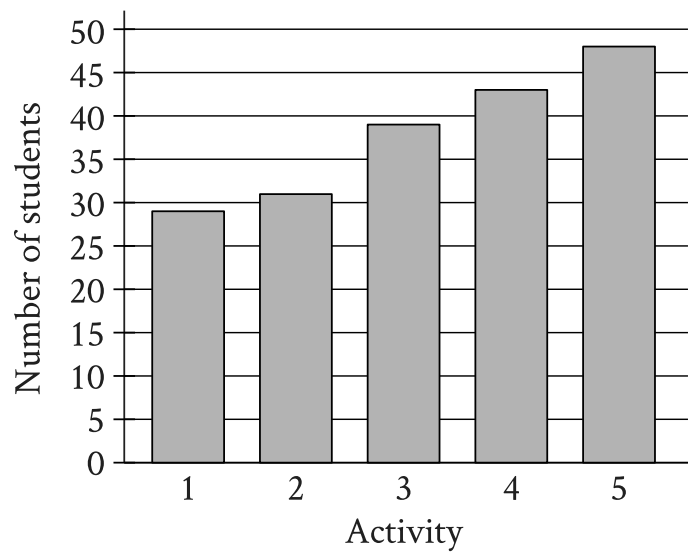
- A. 37
- B. 47
- C. 423
- D. 460

Question ID 93779b53

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 93779b53

SAT3 M 2.2



A group of students voted on five after-school activities. The bar graph shows the number of students who voted for each of the five activities. How many students chose activity **3**?

- A. **25**
- B. **39**
- C. **48**
- D. **50**

Question ID 349a5bc1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 349a5bc1

SAT3 M 2.3

$4x + 5 = 165$

What is the solution to the given equation?

Question ID 86636d8f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 86636d8f

SAT3 M 2.4

A customer spent **\$27** to purchase oranges at **\$3** per pound. How many pounds of oranges did the customer purchase?

Question ID 0d391910

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 0d391910

SAT3 M 2.5

The function f is defined by $f(x) = 4x$. For what value of x does $f(x) = 8$?

Question ID 3ea87153

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 3ea87153

SAT3 M 2.6

The function g is defined by $g(x) = x^2 + 9$. For which value of x is $g(x) = 25$?

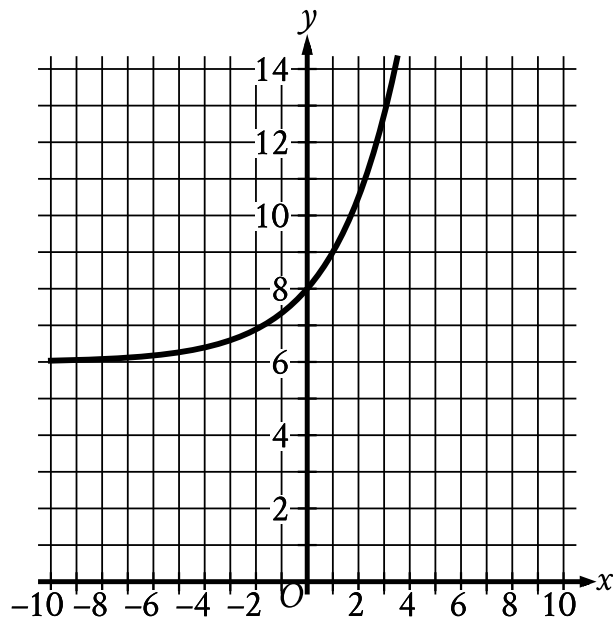
- A. 4
- B. 5
- C. 9
- D. 13

Question ID f547a8b1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: f547a8b1

SAT3 M 2.7



What is the y -intercept of the graph shown?

- A. $(-8, 0)$
- B. $(-6, 0)$
- C. $(0, 6)$
- D. $(0, 8)$

Question ID b8cbe394

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: b8cbe394

SAT3 M 2.8

Sean rents a tent at a cost of **\$11** per day plus a onetime insurance fee of **\$10**. Which equation represents the total cost c , in dollars, to rent the tent with insurance for d days?

- A. $c = 11(d + 10)$
- B. $c = 10(d + 11)$
- C. $c = 11d + 10$
- D. $c = 10d + 11$

Question ID 433184f1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	■ ■ ■

ID: 433184f1

SAT3 M 2.9

Which expression is equivalent to $\frac{4}{4x-5} - \frac{1}{x+1}$?

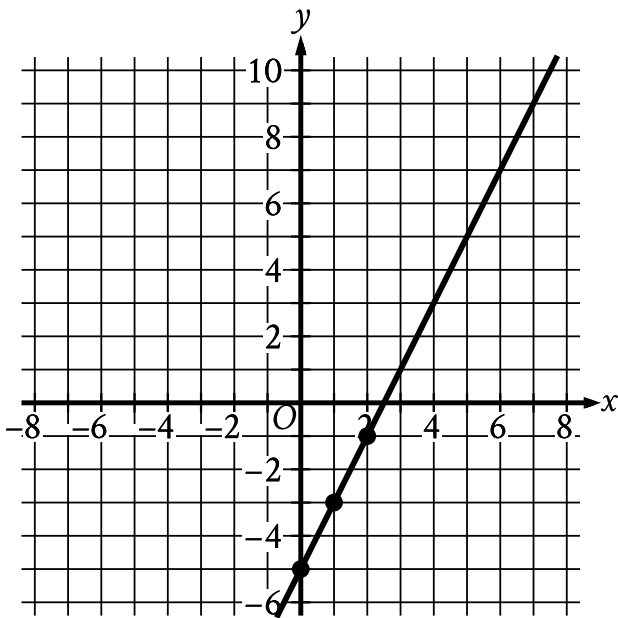
- A. $\frac{1}{(x+1)(4x-5)}$
- B. $\frac{3}{3x-6}$
- C. $-\frac{1}{(x+1)(4x-5)}$
- D. $\frac{9}{(x+1)(4x-5)}$

Question ID 4acd05cd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: 4acd05cd

SAT3 M 2.10



The graph shows the linear relationship between x and y . Which table gives three values of x and their corresponding values of y for this relationship?

A.

x	y
0	0
1	-7
2	-9

C.

x	y
0	-5
1	-7
2	-9

B.

x	y
0	0
1	-3
2	-1

D.

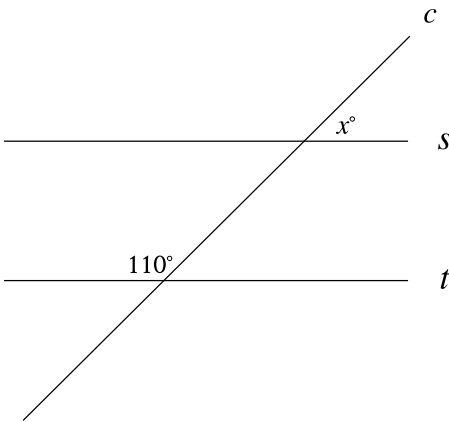
x	y
0	-5
1	-3
2	-1

Question ID cf0d3050

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: cf0d3050

SAT3 M 2.11



Note: Figure not drawn to scale.

In the figure shown, line *c* intersects parallel lines *s* and *t*. What is the value of *x*?

Question ID 02b02213

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 02b02213

SAT3 M 2.12

What is the perimeter, in inches, of a rectangle with a length of **4** inches and a width of **9** inches?

- A. **13**
- B. **17**
- C. **22**
- D. **26**

Question ID b8c4a1cd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: b8c4a1cd

SAT3 M 2.13

$8j = k + 15m$

The given equation relates the distinct positive numbers j , k , and m . Which equation correctly expresses j in terms of k and m ?

- A. $j = \frac{k}{8} + 15m$
- B. $j = k + \frac{15m}{8}$
- C. $j = 8(k + 15m)$
- D. $j = \frac{k+15m}{8}$

Question ID cfe67646

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: cfe67646

SAT3 M 2.14

The point $(8, 2)$ in the xy -plane is a solution to which of the following systems of inequalities?

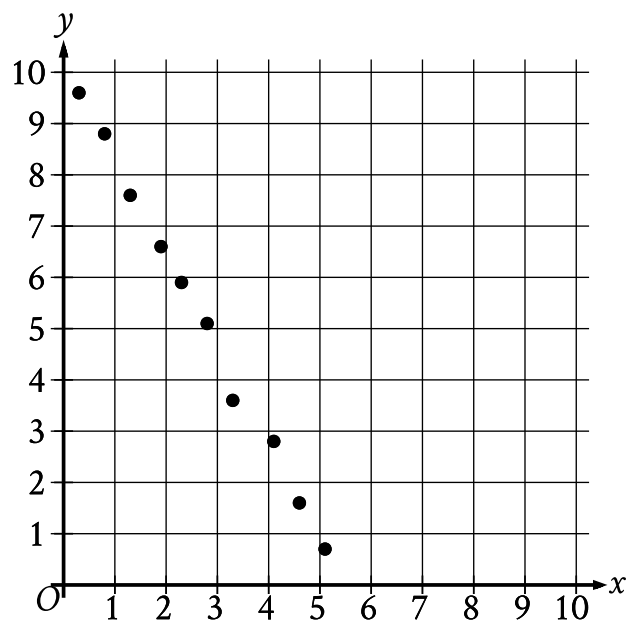
- A. $x > 0$
 $y > 0$
- B. $x > 0$
 $y < 0$
- C. $x < 0$
 $y > 0$
- D. $x < 0$
 $y < 0$

Question ID 5f3ee607

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 5f3ee607

SAT3 M 2.15



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

- A. $y = -1.9x - 10.1$
- B. $y = -1.9x + 10.1$
- C. $y = 1.9x - 10.1$
- D. $y = 1.9x + 10.1$

Question ID d8ace155

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: d8ace155

SAT3 M 2.16

A company opens an account with an initial balance of **\$36,100.00**. The account earns interest, and no additional deposits or withdrawals are made. The account balance is given by an exponential function **A** , where **$A(t)$** is the account balance, in dollars, **t** years after the account is opened. The account balance after **13** years is **\$68,071.93**. Which equation could define **A** ?

- A. $A(t) = 36,100.00(1.05)^t$
- B. $A(t) = 31,971.93(1.05)^t$
- C. $A(t) = 31,971.93(0.05)^t$
- D. $A(t) = 36,100.00(0.05)^t$

Question ID 3a9d60b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■ ■ ■

ID: 3a9d60b2

SAT3 M 2.17

$2|4 - x| + 3|4 - x| = 25$

What is the positive solution to the given equation?

Question ID b4a6ed81

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: b4a6ed81

SAT3 M 2.18

The expression $90y^5 - 54y^4$ is equivalent to $ry^4(15y - 9)$, where r is a constant. What is the value of r ?

Question ID 4993b828

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 4993b828

SAT3 M 2.19

The area A , in square centimeters, of a rectangular cutting board can be represented by the expression $w(w + 9)$, where w is the width, in centimeters, of the cutting board. Which expression represents the length, in centimeters, of the cutting board?

- A. $w(w + 9)$
- B. w
- C. 9
- D. $(w + 9)$

Question ID 48fb34c8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	■ ■ ■

ID: 48fb34c8

SAT3 M 2.20

$$y > 13x - 18$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	21
5	47
8	86

B.

x	y
3	26
5	42
8	86

C.

x	y
3	16
5	42
8	81

D.

x	y
3	26
5	52
8	91

Question ID cd358b89

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: cd358b89

SAT3 M 2.21

Function f is defined by $f(x) = (x + 6)(x + 5)(x + 1)$. Function g is defined by $g(x) = f(x - 1)$. The graph of $y = g(x)$ in the xy -plane has x -intercepts at $(a, 0)$, $(b, 0)$, and $(c, 0)$, where a , b , and c are distinct constants. What is the value of $a + b + c$?

- A. -15
- B. -9
- C. 11
- D. 15

Question ID ae041e52

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	■ ■ ■

ID: ae041e52

SAT3 M 2.22

A square is inscribed in a circle. The radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. What is the side length, in inches, of the square?

- A. 20
- B. $\frac{20\sqrt{2}}{2}$
- C. $20\sqrt{2}$
- D. 40